Global-margin conditioning at the Gaussian endpoint: normalization, training loss, and output amplitude

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Abstract

We study the exact Gaussian-process posterior obtained by imposing a fixed absolute output margin at every Boolean input and then observing noiseless training signs. This is the $\gamma = 0$ width limit of a specified Gaussian tanh network prior. For fixed depth, a random sample of size $n \le e^{d/25}$ has, simultaneously for every labeling, a raw conditioning probability at least $\exp[-C(d^2 + nd + D_r \log d)]$, where $r \sim d/200$ and D_r is the dimension of Boolean polynomials of degree at most r. Consequently the actual latent posterior has average training log loss $O(d)$ when $n \ge D_r \log d$. We also prove that a posterior output draw has magnitude at least $cd^{(1-\beta)/2}$ at all but an $O(1/d)$ fraction of inputs in expectation, where $\beta = 2^{-(L-1)}$. The proofs retain every global margin constraint. They use a comparison covariance, localized interpolation, and independent auxiliary subsets. These are conditioning estimates, not an exponential sample-complexity lower bound or an identification of posterior confidence with mean prediction. All kernel estimates needed for the conclusions are included.

1 Model and results

Let $\mathcal{X} = \{-1, 1\}^d$, equipped with uniform probability, and $M = 2^d$. Fix a depth $L \ge 2$ and an absolute margin $a > 0$. Define

$$\Psi(t) = \frac{2}{\pi} \arcsin t, \qquad F = \Psi^{\circ(L-1)}, \qquad K(x, y) = F\left(\frac{1 + x \cdot y}{d + 1}\right), \qquad \beta = 2^{-(L-1)}.$$

For $G \sim N(0, K)$, inputs $S = (X_1, \dots, X_n)$, and labels $y_i \in \{-1, 1\}$, put

$$A_S = \{|G(x)| > a \ \forall x \in \mathcal{X};\ y_i G(X_i) > 0 \ \forall i\}, \qquad D(S) = -\log \mathbb{P}(A_S).$$

The posterior throughout this paper is the law of G conditional on A_S. This is exactly the same law as first conditioning on the global margin and then on training signs. The probability in $D(S)$ is taken under the original, unconditioned Gaussian law.

The Walsh character $\chi_U(x) = \prod_{i \in U} x_i$ is an eigenfunction of the normalized covariance operator K/M. Its eigenvalue is denoted by $\lambda_{|U|}$. For an odd cutoff r let $D_r = \sum_{j \le r} \binom{d}{j}$ and let τ_r be the diagonal of the covariance retaining degrees greater than r. Write $v = \lambda_{\min}(K) > 0$. The distinction between the matrix eigenvalue v and the normalized eigenvalues λ_j is essential.

There is an exact splitting

$$G = H + \sqrt{v}\, \xi, \qquad H \sim N(0, K - vI_M), \quad \xi \sim N(0, I_M),$$

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with independence. The latent covariance may be singular. Write ρ_S for the actual conditional law of H given A_S, and define

$$B_a(t) = \Phi\left(\frac{t-a}{\sqrt{v}}\right) + \Phi\left(\frac{-t-a}{\sqrt{v}}\right), \quad p_a(t) = \frac{\Phi((t-a)/\sqrt{v})}{B_a(t)}, \quad \ell_a(t) = -\log p_a(t).$$

The quantity $p_a(t)$ is the probability of a positive output conditional on latent value t and on its absolute margin.

Theorem 1.1. *Fix $L \geq 2$ and $a > 0$, and let r be the largest odd integer not exceeding $d/200$. There are positive constants C, c, b, depending only on L, a, such that the following holds for sufficiently large d. For every integer $1 \leq n \leq e^{d/25}$, an independent uniform input sample has probability at least $1 - e^{-cd}$ of satisfying, simultaneously for every labeling,*

$$D(S) \leq C(d^2 + nd + D_r \log d), \tag{1}$$

$$\mathbb{E}_{\rho_S} \frac{1}{n} \sum_{i=1}^{n} \ell_a(y_i H(X_i)) \leq C\left(d + \frac{d^2 + D_r \log d}{n}\right), \tag{2}$$

$$\mathbb{E}_{\rho_S} \frac{1}{n} \sum_{i=1}^{n} (y_i H(X_i))_-^2 \leq Cv\left(d + \frac{d^2 + D_r \log d}{n}\right), \tag{3}$$

$$\mathbb{E}\Big[\mathbb{P}_x\{|G(x)| \leq bd^{(1-\beta)/2}\} \mid A_S\Big] \leq \frac{C}{d}. \tag{4}$$

Here $t_- = \max\{-t, 0\}$. Each specified sample size has the stated probability guarantee; no simultaneous assertion over all prefixes of one infinite sample is intended.

Since $D_r \log d$ eventually exceeds d^2, (2) is $O(d)$ when $n \geq D_r \log d$. Estimate (4) concerns the actual output draw, averaged over the posterior and a uniform input. It does not assert a bound at every input for every posterior draw.

1.1 Relation to the network convention

For completeness, consider width-N networks

$$h^0(x) = (1, x), \quad h^s(x) = \tanh(W^s h^{s-1}(x)) \ (1 \leq s < L), \quad z_N(x) = W^L h^{L-1}(x),$$

with independent centered Gaussian entries and variances

$$\operatorname{Var} W_{ij}^1 = N/d, \qquad \operatorname{Var} W_{ij}^s = 1 \ (1 < s < L), \qquad \operatorname{Var} W_i^L = N^{-1}.$$

Equivalently, the normalized weight scale follows the prescribed schedule $t(N) = \sqrt{N}$: the three variances are $t(N)^2/d$, $t(N)^2/N$, and $t(N)^2/N^2$. These are the $\gamma = 0$ scalings. At fixed d, L, the empirical hidden-layer covariances converge, by conditional laws of large numbers and induction, to iterates of Ψ. Indeed preactivation variances diverge, tanh converges to the sign function under a Gaussian coupling, and the Gaussian sign-correlation identity is $\mathbb{E} \operatorname{sign} U \operatorname{sign} V = (2/\pi) \arcsin \rho$. Continuous matrix square roots and bounded convergence justify the conditional covariance limit, including singular intermediate matrices. The first layer normalizes the augmented inputs to correlation $(1 + x \cdot y)/(d + 1)$. The final output is conditionally Gaussian with the empirical last-hidden covariance, so $z_N \Rightarrow G$ as a vector on the finite cube.

The spectral proof in Appendix A shows K is positive definite. Every boundary $G(x) = \pm a$ or $G(x) = 0$ is a null set, and every compatible open margin-and-label event has positive probability. Ratios of probabilities therefore show convergence to the posterior used here when $N \to \infty$ at fixed d, L, a, S. No interchange of width and dimension limits is claimed.

Proof roadmap. Localized kernel functions interpolate arbitrary labels while maintaining a margin on the whole cube. A damped low-degree comparison covariance turns this construction into a lower bound on the conditioning probability. The exact latent density then gives the training-loss bounds. Finally, independent auxiliary input sets convert the same conditioning cost into an amplitude bound for actual output draws. The appendices supply the spectral and random covariance estimates.

2 A comparison covariance and a global interpolation bound

For a covariance J on the cube, use the Gaussian Cameron–Martin norm $\|h\|_J^2 = h^T J^{-1} h$. For a translation-invariant kernel it equals $\sum_U \widehat{h}(U)^2/\lambda_{|U|}^J$, using normalized Walsh coefficients.

Let Q retain degrees greater than r, let $K_{\mathrm{low}} = K - Q$, and set $\tau = \tau_r$. Define

$$\widetilde{K} = \tau K_{\mathrm{low}} + Q.$$

Appendix A proves $\tau \asymp d^{-\beta}$, $\lambda_0 \geq c/d$, and $v \geq c(d \log d)^{-\beta}$. In particular $0 < \tau < 1$ for large d. The diagonal of $\widetilde{K}$ is $\tau(1-\tau) + \tau \leq 2\tau$. Its low-degree eigenvalues are those of K multiplied by τ, while all remaining eigenvalues agree. The Gaussian densities thus satisfy, pointwise,

$$\frac{dN(0,K)}{dN(0,\widetilde{K})}(g) = \tau^{D_r/2} \exp\left[\frac{1}{2} g^T (\widetilde{K}^{-1} - K^{-1}) g\right] \geq \tau^{D_r/2}. \tag{5}$$

The comparison covariance is only a proof device; it does not replace the prior in the theorem.

Lemma 2.1 (Localized interpolation)**.** *For all sufficiently large d, with probability at least $1 - e^{-cd}$ over the sample in Theorem 1.1, simultaneously for every labeling and every $A > 0$, there is a function h such that*

$$|h(x)| > A \quad \forall x, \qquad y_i h(X_i) > A \quad \forall i, \qquad \|h\|_{\widetilde{K}}^2 \leq CA^2(d/\tau + nd^\beta).$$

Proof. For $z \in \mathcal{X}$ define

$$b_z(x) = 20 \int_{2/5}^{9/20} (1-2p)^{\mathrm{dist}(x,z)}\, dp.$$

Then $b_z(z) = 1$, $0 \leq b_z(x) \leq (1/5)^{\mathrm{dist}(x,z)}$, and its Walsh coefficients have magnitudes

$$\widehat{b}_j = 20 \int_{2/5}^{9/20} (1-p)^{d-j} p^j\, dp, \qquad \binom{d}{j} \widehat{b}_j \leq \frac{20}{d+1}, \qquad \widehat{b}_j \leq (3/5)^d.$$

The middle inequality follows by extending to the complete beta integral. Appendix A gives

$$\lambda_j \geq \frac{cd^{-1-\beta}}{\binom{d}{j}} \quad (j \leq .49d).$$

Therefore the contribution of these degrees to $\|b_z\|_K^2$ is at most

$$Cd^{1+\beta} \sum_{j \leq .49d} \left[\binom{d}{j} \widehat{b}_j\right]^2 \leq Cd^\beta.$$

For $j > .49d$ the integrand increases with p on the integration interval, giving $\widehat{b}_j \le .55^{d-j}.45^j$. Since every $\lambda_j \ge v/M$, the remaining energy is at most

$$\frac{2^d}{v} \sum_{j>.49d} \binom{d}{j} .55^{2(d-j)}.45^{2j} \le \mathrm{poly}(d) e^{-c_* d},$$

where

$$c_* = -[\log 2 + h(.49) + 1.02\log(.55) + .98\log(.45)] > .0062.$$

Here $h(t) = -t\log t - (1-t)\log(1-t)$, and the binomial entropy bound applies because the exponent decreases for $j/d \ge .49$.

The low-degree energy increases by at most

$$\tau^{-1} D_r (.72)^d / v$$

under damping. Since $D_r \le \exp[(h(1/200) + o(1))d]$ and $h(1/200) < .04$, this term decays exponentially. Hence $A_d := \|b_z\|^2_{\widetilde{K}} \le Cd^\beta$.

Let $J(z,w) = \langle b_z, b_w \rangle_{\widetilde{K}}$. Its normalized Fourier eigenvalues are $\nu_j = \widehat{b}_j^2 / \lambda_j^{\widetilde{K}} \le .72^d/(\tau v)$. Parseval and Cauchy–Schwarz give

$$\mathbb{E}_{X,Y} |J(X,Y)| \le \left(\sum_j \binom{d}{j} \nu_j^2 \right)^{1/2} \le \sqrt{\nu_{\max} A_d}.$$

For every subset T of the sampled indices, even one chosen after seeing the sample,

$$\left\| \sum_{i \in T} b_{X_i} \right\|^2_{\widetilde{K}} \le n A_d + \sum_{i \ne j} |J(X_i, X_j)|.$$

The expected cross sum is at most $n(n-1)\sqrt{\nu_{\max} A_d}$. Markov's inequality at threshold nd^β has exponentially small failure when $n \le e^{.04d}$, because $\frac{1}{2}\log(1/.72) > .16$. Thus all subsets simultaneously have energy at most Cnd^β.

Also, with exponentially high probability, every pair of different sample indices has inputs at distance greater than $.2d$: a union bound costs at most $n^2 \exp[-d(\log 2 - h(.2))]$, with $\log 2 - h(.2) > .19$. This event excludes repetitions. At any input at most one center is within distance $.1d$, and the contribution of all other centers is at most $n(.2)^{.1d} = o(1)$. Thus the bump sum over any selected centers is $1 + o(1)$ at a selected center and at most $.2 + o(1)$ everywhere else.

For $T = \{i : y_i = -1\}$ take $h = 2A(1 - 2\sum_{i \in T} b_{X_i})$. The preceding bounds give its strict global and training margins. Since $\|1\|^2_{\widetilde{K}} = 1/(\tau\lambda_0) \le Cd/\tau$, the squared triangle inequality gives the asserted energy bound. □

Proof of (1). Put $t_0 = \sqrt{4\tau \log(4M)}$ and $A = a + t_0 + 1$. Under $N(0, \widetilde{K})$ the symmetric event $B = \{\max_x |G(x)| \le t_0\}$ has probability at least $1/2$, by the diagonal bound and a Gaussian union bound. Lemma 2.1 makes $h + B$ a subset of A_S. Because a is fixed, $\tau \asymp d^{-\beta}$, and $\beta < 1$, we have $A^2 \le C_a \tau d$ for large d. Hence

$$\|h\|^2_{\widetilde{K}} \le C(d^2 + nd).$$

For a symmetric event B, the Gaussian translation formula, symmetry, and $\cosh t \ge 1$ imply

$$\mathbb{P}_{\widetilde{K}}(h + B) \ge \mathbb{P}_{\widetilde{K}}(B) e^{-\|h\|^2_{\widetilde{K}}/2}.$$

Combine this with (5) and $\log(1/\tau) = O(\log d)$ to obtain (1). □

3 The exact posterior training loss

On the sample event in Lemma 2.1, all observed inputs are distinct. Conditional independence of the noise coordinates in $G = H + \sqrt{v}\xi$ gives the exact latent posterior density

$$d\rho_S(H) = \frac{\exp[-V(H) - \sum_i \ell_a(y_i H(X_i))]}{\mathbb{P}(A_S)}\, d\nu(H), \qquad V(H) = -\sum_{x\in\mathcal{X}} \log B_a(H(x)),$$

where $\nu = N(0, K - vI_M)$. All terms in the exponent are nonnegative. The Gibbs identity is consequently

$$D(S) = \mathrm{KL}(\rho_S \| \nu) + \mathbb{E}_{\rho_S} V(H) + \mathbb{E}_{\rho_S} \sum_i \ell_a(y_i H(X_i)). \tag{6}$$

This holds on the support of ν even when its covariance is singular. The density is bounded above by $e^{D(S)}$, which also justifies integrability and the identity. Nonnegativity proves (2).

For completeness, $\ell_a(t) \geq (t_-)^2/(2v)$. To see this write $t = -u \leq 0$ and $\overline{\Phi} = 1 - \Phi$. If $0 \leq u \leq a$, the normal-tail ratio bound

$$\frac{\overline{\Phi}(A)}{\overline{\Phi}(B)} \leq e^{-(A^2-B^2)/2} \qquad (A \geq B \geq 0)$$

implies $p_a(-u) \leq e^{-2au/v}$. The ratio bound follows by integrating the normal hazard $\phi(t)/\overline{\Phi}(t) \geq t$. If $u \geq a$, the denominator in $p_a(-u)$ is at least $1/2$ and $\overline{\Phi}(t) \leq \frac{1}{2}e^{-t^2/2}$ for $t \geq 0$, giving $p_a(-u) \leq e^{-(a+u)^2/(2v)}$. Both cases imply the claimed inequality; for $t \geq 0$ it is immediate. Equation (6) now proves (3).

4 Auxiliary subsets force large actual output magnitudes

The next lemma is independent of the cube and of the training likelihood.

Lemma 4.1 (A finite Gaussian amplitude estimate)**.** *Let G be a nondegenerate centered Gaussian vector on a finite set. Suppose an event A implies $|G(x)| > a$ at every coordinate and $-\log \mathbb{P}(A) \leq D$, with $D \geq 1$. Let $T = (x_1, \ldots, x_m)$ be distinct coordinates with $K_T \succeq wI_m$, where $w > 0$. Assume $D < m$ and $e^{-2a^2/w} \leq 1/2$. Set*

$$t = \sqrt{\log(m/D)}, \qquad b_0 = a + \sqrt{w}\, t, \qquad c_0 = \frac{e^{-3/2}}{\sqrt{2\pi}}.$$

Then

$$\mathbb{E}\left[\frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{|G(x_i)|\leq b_0/2\}} \,\middle|\, A\right] \leq \frac{2}{c_0}(t+1)\frac{D+1}{\sqrt{mD}} + \frac{8w}{b_0^2} + \frac{16wD}{mb_0^2}. \tag{7}$$

The event may impose arbitrary further constraints outside T.

Proof. On an enlarged Gaussian space draw $H_T \sim N(0, K_T - wI)$ and $\xi \sim N(0, I_m)$ independently, set $G_T = H_T + \sqrt{w}\xi$, and draw the other coordinates from their original conditional law given

G_T. This reproduces the original full law of G. The outside coordinates may depend on ξ; no independence from them is asserted. The event inclusion alone gives

$$\mathbb{P}(A \mid H_T) \le \prod_{i=1}^{m} B_{a,w}(H_T(i)), \quad B_{a,w}(h) = \Phi((h-a)/\sqrt{w}) + \Phi((-h-a)/\sqrt{w}).$$

The even function $B_{a,w}$ increases with $|h|$. If $q_H = m^{-1}\#\{i : |H_T(i)| \le b_0\}$ and $c_b = -\log B_{a,w}(b_0)$, then

$$\mathbb{P}(q_H \ge q \mid A) \le \min\{1, e^{D - mc_b q}\}, \qquad \mathbb{E}[q_H \mid A] \le \frac{D+1}{mc_b}.$$

Now $1 - B_{a,w}(b_0) = \overline{\Phi}(t) - \overline{\Phi}(t + 2a/\sqrt{w}) \ge \overline{\Phi}(t)/2$. The ratio bound used above proves this last inequality. Integrating the normal density over $[t, t + 1/(t+1)]$ gives $\overline{\Phi}(t) \ge c_0 e^{-t^2/2}/(t+1)$. Since $-\log B \ge 1 - B$, these estimates give the first term on the right of (7) as an upper bound for $\mathbb{E}[q_H \mid A]$.

The enlarged posterior has relative entropy exactly $-\log \mathbb{P}(A) \le D$ from its prior. Data processing gives the same upper bound for the relative entropy of its ξ marginal from $N(0, I_m)$. The entropy variational inequality and the Gaussian moment generating function imply

$$\tfrac{1}{4}\mathbb{E}[\|\xi\|^2 \mid A] \le D + \log \mathbb{E} e^{\|\xi\|^2/4} = D + \tfrac{m}{2}\log 2,$$

so $\mathbb{E}[\|\xi\|^2 \mid A] \le 4D + 2m$. If $|G_T(i)| \le b_0/2$ and $|H_T(i)| > b_0$, then $\sqrt{w}|\xi_i| > b_0/2$. The averaged Markov inequality completes the proof. □

Proof of (4). Hold fixed any training sample on the event proving (1), and any labeling. Increase its deterministic upper bound D to be at least one. Since $n \le e^{.04d}$ and $h(1/200) < .04$, for large d we can take $D \le e^{.045d}$.

Independently draw an auxiliary uniform sample T of size $m = \lfloor e^{.05d} \rfloor$. Let u be odd with $u/d \to 1/50$, and put $w = \tau_u/2 \asymp d^{-\beta}$. Appendix B proves, with exponentially high probability over T,

$$Q_{>u,TT} \succeq (\tau_u/2) I_m,$$

and thus $K_T \succeq w I_m$; this event in particular excludes repeated points. Indeed the bounds there decay exponentially because

$$.05 < h(1/50), \qquad .10 < \log 2 - h(1/4).$$

The far-pair error is exponentially small as well, since u is a fixed positive fraction of d.

Lemma 4.1 applies for every such T, to the same full event A_S. Here $\log(m/D) \ge .004d$ for all sufficiently large d, so $b_0^2 \ge cd^{1-\beta}$; its first error term is exponentially small and the other terms are $O(1/d)$. Choose a fixed positive b with $bd^{(1-\beta)/2} \le b_0/2$.

Although the enlarged splitting in the lemma depends on T, its marginal law of the full G is always the original Gaussian law, and its conditional marginal is always the same actual posterior $G \mid A_S$. Averaging its conclusion over the independent auxiliary sample therefore changes the left side exactly into

$$\mathbb{E}[\mathbb{P}_x\{|G(x)| \le bd^{(1-\beta)/2}\} \mid A_S].$$

Bad auxiliary samples contribute at most their exponentially small probability. This proves (4) and completes Theorem 1.1. □

5 What these estimates do not prove

For parity labels $f(x) = \prod_i x_i$, define the actual confidence $q_S(x) = \mathbb{P}(f(x)G(x) > 0 \mid A_S)$ and the failure fraction $F_S = \mathbb{P}_x\{q_S(x) \le 2/3\}$. Theorem 1.1 does not give a positive lower bound on F_S. Its training-loss bound still needs a valid comparison with population loss for this globally conditioned posterior. Its amplitude estimate gives magnitudes, not signs or correlation with parity.

In particular, a small positive lower bound on the posterior-draw error would be insufficient: if $R_S = \mathbb{E}[\mathbb{P}_x(f(x)G(x) < 0) \mid A_S]$, then $R_S \le 1/3 + (2/3)F_S$. This implication would require $R_S > 1/3$ to force a positive confidence-failure fraction. Nor is the sign of a posterior mean used as a substitute for confidence. The latent posterior contains the full factor $\prod_x B_a(H(x))$; no log-concavity of this posterior is assumed. An exponential sample lower bound is proved separately in the later paper [1], including this globally conditioned Gaussian endpoint. The present theorem remains an auxiliary conditioning estimate; its displayed conclusions alone do not imply that lower bound.

A Fourier level masses of the kernel

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, and $F = \Psi^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j\in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work on the full D-cube with kernel $F(u \cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this and the next section may depend on L. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma A.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m-\beta)}{m!} \asymp m^{-1-\beta} \qquad (m \ge 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of Ψ has positive coefficients at every odd exponent, with

$$[z^{2k+1}]\Psi(z) = \frac{2}{\pi}\frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$\Psi(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^\beta + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [2, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|\Psi(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$.

Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] \\ + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$. Consequently

$$D_r = \sum_{\substack{j \le r \\ j \text{ odd}}} \binom{D}{j}, \quad \tau_r = \sum_{\substack{j > r \\ j \text{ odd}}} \binom{D}{j} a_j, \quad \lambda_{r+1} = a_{r+2}. \tag{8}$$

The odd cutoff is essential for these exact identities.

Lemma A.2 (Sharp level and tail estimates)**.** *Uniformly for odd* $1 \le j \le D/4$ *and odd* $1 \le r \le D/8$,

$$\binom{D}{j} a_j \asymp j^{-1-\beta}, \tag{9}$$

$$\tau_r \asymp r^{-\beta}, \tag{10}$$

$$\frac{\lambda_{r+1}}{\tau_r} \asymp \frac{1}{r\binom{D}{r+2}}, \tag{11}$$

$$s_{2,r} := \sum_{q > r} \binom{d}{q} \lambda_q^2 \le \frac{C r^{-2\beta-2}}{\binom{D}{r+2}}. \tag{12}$$

The same level-mass estimate holds up to $j \le \rho D$ *for any fixed* $\rho < 1/2$*, with constants depending on* ρ*.*

Proof. The gamma integral and Lemma A.1 give, for odd j,

$$a_j \asymp D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \quad p(u) = \frac{1-e^{-2u}}{2}, \tag{13}$$

$$\asymp D^{-\beta} \int_0^{1/2} p^j (1-p)^{D-j} w_\beta(p)\, dp, \tag{14}$$

where $w_\beta(p) = [-\frac{1}{2}\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0,b]$ for every fixed $b < 1/2$.

Fix $\rho < b < 1/2$ and take $j \le \rho D$. The integral on $[0,b]$ is comparable to the complete beta integral $B(j-\beta, D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1-\rho/b$ of its mass in $[0,b]$. The identity

$$\binom{D}{j} B(j-\beta, D-j+1) = \frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)} \asymp D^\beta j^{-1-\beta}$$

gives the desired mass. On $[b, 1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (9) and its stated extended range.

Sum (13) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r \le CD^{-\beta} \int_0^\infty u^{-1-\beta} \min\{1, Du/(r+1)\}\, du \le C(r+1)^{-\beta}.$$

For the lower bound, sum (9) over the odd j in $(r, 2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (11) follows from (9), (10), and (8).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j = r+2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (12). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{15}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

Corollary A.3 (Lower bounds used by the interpolation)**.** *For sufficiently large d,*

$$\lambda_0 \ge c/d, \qquad \lambda_j \ge \frac{cd^{-1-\beta}}{\binom{d}{j}} \quad (0 \le j \le .49d), \qquad v := \lambda_{\min}(K) \ge c(d\log d)^{-\beta}.$$

Also $v \to 0$.

Proof. The first bound follows from the level estimate at full-cube degree one. For odd reduced degree j, use $\lambda_j = a_j$ and $\binom{D}{j}/\binom{d}{j} = D/(D-j)$. For even j, use $\lambda_j = a_{j+1}$ and $\binom{D}{j+1}/\binom{d}{j} = D/(j+1)$. The extended level bound with any fixed cutoff between .49 and 1/2 then gives the stated uniform lower bound, including $j = 0$.

For any odd full-cube degree $j \le D$, restrict the integral (13) to $u \ge \frac{1}{2}\log(2D)$. In this range $p(u) \ge (1 - 1/(2D))/2$ and $1 - p(u) \ge 1/2$, so $p(u)^j(1-p(u))^{D-j} \ge c2^{-D}$. Integrating $u^{-1-\beta}$ gives $a_j \ge c2^{-D}D^{-\beta}(\log D)^{-\beta}$. Every reduced eigenvalue equals an odd a_j, and the matrix eigenvalues are M times the normalized ones. Thus $v \ge c(d\log d)^{-\beta}$. Finally, for any fixed small positive cutoff fraction, the minimum normalized eigenvalue is at most the average in its tail, whence $v \le M\tau_r/(M - D_r) = O(d^{-\beta})$. This proves $v \to 0$. □

B Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma B.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \le D/8$ is odd,*

$$\max_{d/4 \le \mathrm{dist}(x,y) \le 3d/4} |Q(x,y)| \le \rho_r := C\sqrt{D}\, e^{-r/8} + Ce^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h;D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \le h \le (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b(1+t)^{D-2b} \le (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2}\right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \le j < D$,

$$\frac{|\mathsf{K}_j(h;D)|}{\binom{D}{j}} \le C\sqrt{D}\exp[-2\delta j(1-j/D)]. \tag{16}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j>r,\ j \text{ odd}} a_j \mathsf{K}_j(h;D)$. For $r < j \le D - r$, (16) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \le a_{b_D} \sum_{k<r} \binom{D}{k} \le CD^{-\beta-1/2}\exp[-D(h(1/4) - h(1/8)) + O(1)].$$

The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma B.2 (A tail Gram deviation bound)**.** *Let $X_1, \dots, X_n$ be independent uniform cube points, with $1 \ll n \le M/4$. Let*

$$B_{\mathrm{near}} = \#\{y : \mathrm{dist}(x, y) \notin [d/4, 3d/4]\} \le 2e^{dh(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\begin{aligned}&\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\mathrm{op}} > z\tau\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\rho_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1 + B_{\mathrm{near}})}{M}\right]^2 + \frac{n^2(1 + B_{\mathrm{near}})}{M}. \end{aligned} \tag{17}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\begin{aligned}\|J_{\mathrm{far}}\|_{\max} &\le \rho_r, \\ \|J_{\mathrm{far}}\|_{1\to 2}^2 &\le Ms_2, \\ \|J_{\mathrm{far}}\|_{\mathrm{op}} &\le M\ell + \tau + \tau B_{\mathrm{near}}.\end{aligned}$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [3, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\mathrm{far}}R\|^2)^{1/2} \le C[q\rho_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1 + B_{\mathrm{near}})]. \tag{18}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \mathrm{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (18) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\mathrm{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (17). □

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